EMERGENCY FIRST-RESPONDER PLATFORM

SANJEEVANI

Practice before the moment.

An on-device, AR-assisted emergency-skills training system transforming bystander CPR and crisis protocols through real-time computer vision.



♡ CPR Protocol ★ Stroke FAST ♢ Heart Attack ⊕ Allergic Protocol

Android · MediaPipe On-Device ML · Zero Cloud Latency · Fully Offline

The Moment Before Help Arrives

The critical gap between theoretical awareness and procedural execution

"Imagine someone sitting next to you suddenly collapses. Emergency services (112) have been alerted and are en route. What will you actually do with your hands during the next six minutes?"

Most People Know About Emergencies

- Seen CPR depicted in cinema and television
- Watched online awareness or training clips
- Understand that dialing 112 is the immediate first step

Declarative Knowledge: Knows the concept, but has never calibrated hands, tempo, or posture under stress.

Far Fewer Have Practised Responding

- Hands positioned accurately over lower sternum
- Maintained 100–120 compressions/min without fatigue
- Received live correction for locked elbows or drift

Procedural Confidence: Muscle memory built through repeated, verified attempts with instant feedback.

"The problem is not lack of awareness. It is the absence of accessible, repeatable practice."

The Access Chasm in India

Four systemic friction points limiting hands-on emergency training

1. Qualified Instructor Scarcity

Certified BLS instructors are scarce and clustered in metro areas.

Courses demand scheduling, venue booking, travel, and trainer hours.

2. Passive Learning Inertia

Lectures and videos explain concepts, but cannot observe learners.

Cannot detect misplaced wrists, inadequate tempo, or arm fatigue.

3. Dedicated Hardware Costs

Feedback sensor manikins represent a meaningful procurement barrier.

Capital costs, storage, calibration, and replacement limit cohort size.

4. Rapid Skill Decay Over Time

Motor skills decline sharply within 3 to 6 months without drills.

One-off annual workshops fail; retention requires frequent repetition.

"Information is already ubiquitous. Scalable opportunities for interactive revision are not."

A Video Explains the Steps. Can It Correct Your Form?

The biomechanical divide between passive observation and interactive correction

PASSIVE LEARNING [One-Way]

- Video demonstrations and recorded webinars
- Classroom slide decks and printed posters
- Mass community awareness seminars

Educational Value:

Clarifies sequencing and emergency concepts.

Critical Limitation:

Zero observation; cannot tell if hands are 5 cm off target.

VS

INTERACTIVE PRACTICE [Closed-Loop]

- Active physical execution by the learner
- Camera-based spatial landmark tracking
- Real-time AR directional and audio guidance

Biomechanical Value:

Calibrates muscle memory and positional accuracy.

Verified Progress:

Learner discovers errors and refines posture dynamically.

"Watching teaches what the steps are. Practising reveals what you can actually execute."

Physical Equipment is Vital — But Hard to Scale for Revision

Evaluating manikin tiers and the logistics of continuous institutional practice

Training Equipment Capabilities

Hardware Tier	Capabilities & Limits
Basic Manikin	Anatomical surface; no sensor feedback; cannot verify compression rate or recoil.
Feedback Manikin	Sensor measures rate and depth; audio/LED signals; requires dedicated storage.
Advanced QCPR	Full telemetry, AED trainer sync, detailed analytics; high institutional cost.

*Published list prices vary widely by supplier. Advanced systems represent significant capital expenditure for budget-constrained institutions.

Total Cost of Training Logistics

- **Capital Outlay:** Hardware sets for 30+ students
- **Storage & Sanitization:** Dedicated facility care
- **Scheduling Friction:** Bottlenecks around trainer hours
- **Decay Dilemma:** Equipment stored between workshops

The Architectural Question

How do we enable high-frequency corrective practice between formal workshops using hardware trainees already carry?

Hands-On Emergency Training is an Established Global Model

Learning from AHA and ERC standards while designing for Indian scale

What Resuscitation Science Shows

- The **AHA** and **Resuscitation Council UK** demonstrate that immediate bystander CPR doubles or triples cardiac arrest survival.
- Survival gains correlate strictly with **compression quality** (rate 100–120 BPM, central recoil), not abstract awareness [AHA 2020].
- **The Opportunity in India:** Bridge the gap between formal sessions by turning the ubiquity of smartphones into a personal practice mirror.

Comparing Training Delivery Modes

Method	Primary Strength	Friction Point
Video	Universal access	Zero personal feedback
Classroom	Human instruction	Scheduling & scarcity
Manikin	Tactile feel	Equipment logistics
Sanjeevani	**On-demand AR**	**Observational only**

Sanjeevani complements physical workshops by keeping procedural skills active between sessions.

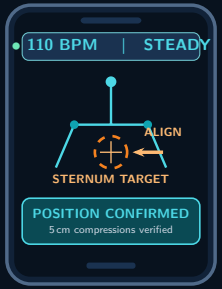
"Sanjeevani adopts the hands-on feedback philosophy of AHA/ERC and delivers it via on-device mobile vision."

Practice That Starts With the Phone in Your Pocket

Introducing Sanjeevani — An on-device Android emergency-skills training platform

Core Platform Capabilities

- ✓ **Interactive Workflows:** Guided emergency crisis drills
- ✓ **Real-Time AR Overlays:** MediaPipe pose and hand tracking
- ✓ **Spoken Guidance:** Low-latency deterministic voice coaching
- ✓ **Deduplicated Telemetry:** Errors grouped into clinical episodes
- ✓ **Targeted Remediation:** Surfaces personalized weak-skill drills
- ✓ **100% Offline AI:** Runs on-device; zero internet required

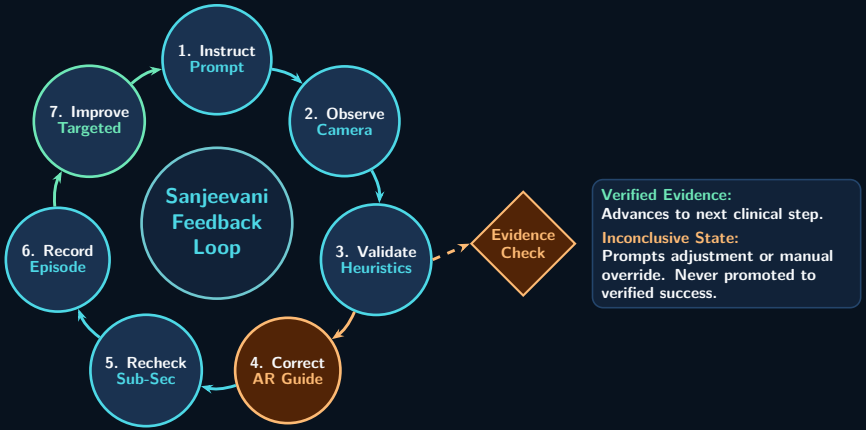


Medical & Engineering Scope:

Sanjeevani is an **educational training aid**, not a diagnostic tool. Engineered to reinforce practice between accredited courses, never to replace certified instructors or manikins.

The 7-Stage Closed Learning Loop

Continuous cycle: instruction, sensor observation, correction, and telemetry



The decision diamond represents honest engineering: when camera tracking is uncertain, the system declares an inconclusive state rather than reporting false mastery.

One Engine. Four Critical Emergency Scenarios.

Unified computer vision architecture tailored for distinct medical protocols

♡ CPR Protocol

[Implemented]

- ✓ Patient posture check
- ✓ AR sternum alignment
- ✓ Directional error arrow
- ✓ Torso motion counter
- ✓ 110 BPM metronome

Boundary:

Depth cannot be derived.

★ Stroke FAST

[Implemented]

- ✓ FAST protocol checklist
- ✓ Face smile asymmetry
- ✓ Arm drift heuristic
- ✓ Speech clarity check
- ✓ Automatic 112 guide

Boundary:

Visual triage heuristic.

♢ Heart Attack

[Implemented]

- ✓ Conscious posture check
- ✓ Sit-down compliance
- ✓ Loosen chest clothing
- ✓ 300 mg aspirin advisory
- ✓ Auto-escalate to CPR

Boundary:

Escalates if unresponsive.

⊕ Allergic Rx

[Implemented]

- ✓ 6-phase anaphylaxis drill
- ✓ Supine posture check
- ✓ Leg elevation heuristic
- ✓ AR thigh target marker
- ✓ 10-sec injection timer

Boundary:

Simulates drill only.

Choking and Recovery Position protocols listed in roadmap. All medical training content subject to qualified clinical review.

See the Learning Loop in Action

Live demonstration on an Android smartphone

Demonstration Walkthrough Steps

1. Launch app → select CPR module → 112 advisory
2. Deliberately misplace hands on camera view
3. Instant AR arrow: **"Move hands 4 cm left"**
4. Correct placement → green confirmation lock
5. Begin compressions → metronome at 110 BPM
6. Session summary generated with criteria review
7. Analytics flags placement as priority weak-skill
8. Tap **Targeted Practice** → dedicated drill launches

Presenter Checklist: ☐ Android device ☐ Screen mirror ☐ Lighting ☐ Demo data



From Observed Errors to Targeted Practice

Connecting computer vision telemetry to personalized skill remediation



Real-World Remediation Example

- Trainee repeatedly misaligns sternum center.
- Engine detects placement accuracy $< 80\%$ threshold.
- Surfaces a 60-second hand alignment drill.
- Trainee repeats solely until verified lock is achieved.

Honest Engineering Principles

✓ Analytics derive strictly from persisted physical attempts.

✓ Continuous error = 1 episode (no artificial FPS inflation).

✓ Manual continuation is never converted to verified proficiency.

✓ Reuses identical computer vision algorithms for drills.

Lowering the Barrier to Practice — Not the Standard of Care

How Sanjeevani fits into the institutional first-aid training ecosystem

Institutional Value Proposition

▶ **High-Frequency Revision:** Enables digital drills between certified manikin sessions without logistics overhead.

▶ **Zero Marginal Cost:** Operates on everyday Android devices.

▶ **Privacy-First On-Device AI:** Zero video sent to remote servers.

▶ **Objective Progress Visibility:** Educators identify cohort weaknesses before formal certification tests.

Deployment Target Environments

• **Schools & Colleges:** Regular student first-aid drills

• **Corporate Workplaces:** Workplace emergency refreshers

• **Community Centers:** Scalable bystander awareness

• **Commercial Transit:** Fleet driver emergency drills

• **Healthcare Trainees:** Basic skill maintenance

What Sanjeevani Does Not Claim:

× Does not replace certified in-person BLS workshops

× Does not confer formal legal resuscitation credentials

× Does not replace physical manikin tactile resistance

Universal Android Accessibility

Works on standard Android smartphones (API 26+).

No internet connection. No account login required.

Measuring Verified Competence — Not Vanity Downloads

Proposed evaluation metrics and pilot study framework

Proposed Evaluation Metrics [Proposed]

- **Protocol Completion:** Clinical adherence score
- **Correction Latency:** Seconds to correct hand alignment
- **BPM Stability:** Percent time within 100–120 target
- **Score Delta:** Skill improvement across repeat drills
- **Self-Efficacy Score:** Pre/post learner confidence

Important Scientific Honesty:

These are proposed evaluation metrics. No formal clinical trial has concluded. Sanjeevani claims verified biomechanical tracking, not empirical clinical outcome data.

Proposed Pilot Study Protocol

Phase 1: Baseline Assessment

Evaluate untrained participants on sensor manikin.

Phase 2: Structured Smartphone Revision

4 Sanjeevani interactive drills over a 2-week period.

Phase 3: Control Arm Comparison

Parallel cohort receives instructional video only.

Phase 4: Blind Evaluator Post-Test

Certified AHA/ERC instructor evaluates physical performance.

Primary Benchmark: Measurable improvement on certified instructor rubric, not internal app metrics alone.

“The goal is not to replace the training room. It is to make practice possible beyond it.”

Real emergencies do not arrive with advance warning. We cannot place a certified first-aid instructor or a sensor-enabled training manikin beside every citizen whenever they wish to practice. But we can turn the smartphones they already carry into accessible, interactive mirrors for life-saving skills.

Sanjeevani bridges the gap between passive awareness and real-world composure — giving bystanders the opportunity to build muscle memory before the moment someone’s life depends on them.

Universal Smartphone Access — **Honest Clinical Engineering** — **Verified Muscle Memory**

References and Evidence Base

Clinical resuscitation guidelines, empirical literature, and technical frameworks

Clinical Resuscitation Standards

[AHA 2020] American Heart Association.
Part 3: Adult Basic and Advanced Life Support: 2020 AHA Guidelines for CPR and ECC. Circulation, 142:S366–S468, 2020.

[ResusUK 2021] Resuscitation Council UK.
Adult Basic Life Support Guidelines, 2021.
resus.org.uk/library/2021-guidelines

[Sasson 2010] Sasson C., Forman J., et al.
Predictors of survival from out-of-hospital cardiac arrest: systematic review and meta-analysis. Circ Cardiovasc Qual Outcomes, 3(1):63–81.

[Panchal 2020] Panchal A.R. et al.

2020 AHA CPR Quality Guidelines. Circulation, 2020.

Empirical Studies & Technology Base

[PMID 28680517] Rao B.K., et al.
Bystander CPR and survival outcomes in India: observational cohort study.

[PMID 34474756] Sharma M., et al.
Bystander CPR awareness, willingness, and barriers in India.

[PMC10568059] Verma P., et al.
Comprehensive review of community bystander CPR in India: the training divide.

[Google MediaPipe] Google LLC.
MediaPipe Pose, Hand, and Face Landmarker On-Device Inference. ai.google.dev, 2024.

[ERSS 112] Ministry of Home Affairs / MoHFW, Govt of India.

Emergency Response Support System — 112 Initiative.

Manikin pricing and institutional equipment costs vary by supplier and region. All clinical protocols should be validated against local emergency medicine directives.